

Exercise Sheet

Task 1 – Dependencies

Why is read after read not a dependency?

Task 2 – Pipelining

a) Determine the dependencies of the following instructions:

```
I1: ADD t0, t0, t1
I2: SUB t1, t0, t2
I3: BNE t3, t1, jump
I4: ADD t4, t2, t3
jump:
I5: LW t4, 24(t0)
I6: SLLI t3, t3, 4
I7: ADD t5, t4, t5
I8: XOR t0, t2, t3
I9: SW t1, 12(t3)
```

- b) How would the program be executed in a 5-stage pipeline without forwarding or branch-prediction if the branch was not taken?
- c) How many NOPs less are needed if forwarding is possible?

Task 3 – Scoreboarding

Our processor consists of one functional unit for addition and one for multiplication. There is no forwarding, therefore an instruction can only access the value computed by another instruction after it has left the “write back”-stage. In what order will the following instructions be executed?

I1: ADD t0, t0, t1
I2: MUL t1, t0, t1
I3: MUL t2, t2, t3
I4: SUB t3, t0, t3
I5: MUL t4, t0, t1
I6: ADD t5, t0, t3
I7: ADD t5, t0, t2

For each instruction that is currently executed, indicate which registers and functional units are currently reserved or in use, and indicate if instructions are executed at the same time.